# Divide Decimals Using an Algorithm

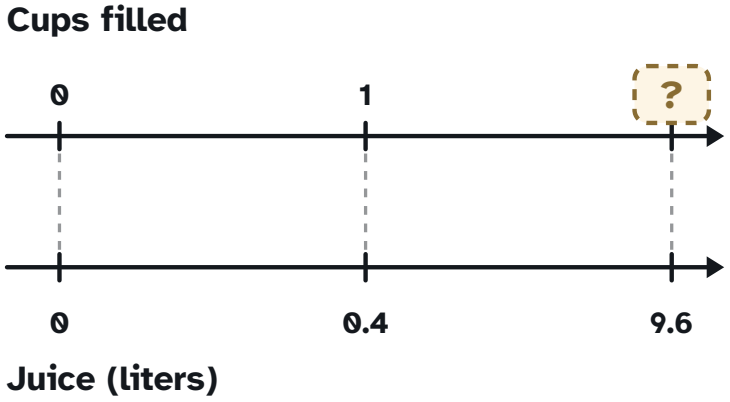
Divide by a decimal by sliding both decimal points the same number of places.

**Name** _____ **Date** _____

## Tonight's spark

A juice stand has 9.6 liters of juice. Each cup holds 0.4 liter. How many cups can it fill?

**Make a guess first: will the number of cups be bigger or smaller than 9.6?**




**Our class model: dividing by a number less than 1**

## Try it

Work it out on paper. Write one answer on each line.

**1** **Sliding a decimal point 1 place to the right multiplies a number by what?**

**A.** 10 **B.** 100 **C.** 2 **D.** 0.1

**My answer** _____

**2** **To make 0.25 a whole number, how far do you slide its decimal point?**

**A.** 1 place to the right, to make 2.5 **B.** 2 places to the right, to make 25

**C.** 3 places to the right, to make 250 **D.** 2 places to the left, to make 0.0025

**My answer** _____

**3** **Which problem has the same answer as 2.4 ÷ 0.6?**

**A.** 2.4 ÷ 6 **B.** 24 ÷ 0.6 **C.** 24 ÷ 6 **D.** 0.24 ÷ 6

**My answer** _____

**4** **What is 3.5 ÷ 0.5?**

**A.** 70 **B.** 0.7 **C.** 3 **D.** 7

**My answer** _____

**5** **What is 4.5 ÷ 0.15?**

**A.** 3 **B.** 30 **C.** 300 **D.** 0.3

**My answer** _____

**6** **How many 0.3-liter cups can be filled from 7.2 liters of water?**

**A.** 2.4 **B.** 240 **C.** 24 **D.** 6.9

**My answer** _____

### Talk about it

Why do both decimal points slide the same number of places?

### Home mission

Pretend a jug holds 9.6 liters and each cup holds 0.4 liter. Before dividing, ask: will there be more cups or fewer cups than 9.6? Why?

### If we get stuck, ask

- Which number is the divisor? Is it whole yet?
- How many places did you move the divisor? Did the dividend move the same?
- We are dividing by less than one — should the answer be bigger or smaller?

**Do it on a screen**

Scan or type the link. Same lesson, with hints.

**eduwonderlab.com/lessons/2-7-part2/homework.html**

**How did it feel? (optional)**

Tricky Getting there Got it